

Verification Report

Enterprise VLSI Chip Design & Verification System

Student: K Murali
Verification Tool: ModelSim
HDL: Verilog

1. Verification Objective

Verify the functionality of each RTL module individually and validate the complete processor through functional simulation.

2. Verification Environment

Simulator: ModelSim
Language: Verilog HDL
Methodology: Module-level and top-level testbench verification.

3. Module Verification Summary

Module	Testbench	Expected Result	Status
Program Counter	program_counter_tb.v	PC increments by 4 and resets to 0	PASS
Instruction Memory	instruction_memory_tb.v	Correct instruction fetched for each PC	PASS
Register File	register_file_tb.v	Read/write operations verified	PASS
ALU	alu_tb.v	Arithmetic and logical operations verified	PASS
Control Unit	control_unit_tb.v	Correct control signals generated	PASS
Data Memory	data_memory_tb.v	LOAD/STORE operations verified	PASS
Processor	processor_tb.v	Integrated processor execution	PASS

4. Test Scenarios

- Reset verification
- Sequential instruction fetch
- Register write/read verification
- ADD, SUB, AND, OR operations
- Memory LOAD and STORE verification
- Write-back path verification

5. Waveform Evidence

Insert ModelSim waveform screenshots for each module and the complete processor here.

6. Verification Results

All individual RTL modules compiled successfully and passed functional simulation. The integrated processor testbench validates interaction between the Program Counter, Instruction Memory, Control Unit, Register File, ALU, and Data Memory.

7. Conclusion

The processor RTL design has been functionally verified using ModelSim. The modular verification approach ensures each component behaves as intended before full processor integration.